Action Angle Variable.

$H_0(J_1, J_2)$ $\qquad$ $\frac{dJ_i}{dt} = -\frac{\partial H_0}{\partial \theta_i} = 0 \Rightarrow J_i = \text{const.}$ $\quad$ action

$$\frac{d\theta_i}{dt} = \frac{\partial H_0}{\partial J_i} = w_i(\vec{J})$$

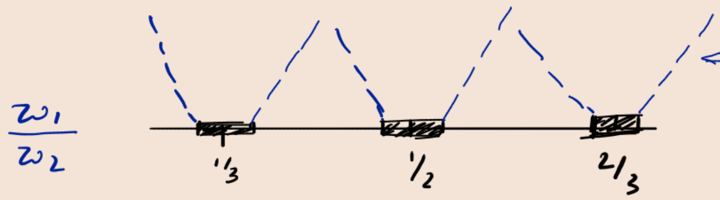
Books on classical mechanics.

Kolmogorov - Arnold Moser Theorem. (KAM Theorem)

If $\det \left| \frac{\partial w_i}{\partial J_j} \right| \neq 0$, then the tori with w_1/w_2 sufficiently irrational, survives if $\exists$ prime integers (m, s) s.t.

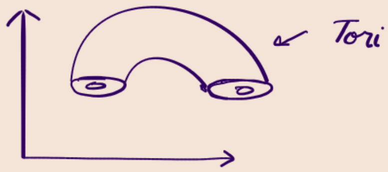
the tori is stable under Perturbations

$$\left| \frac{w_1}{w_2} - \frac{m}{s} \right| > \frac{K(\epsilon)}{s^{5/2}} \qquad \text{for } \epsilon << 1$$



Growth of regions with increasing epsilon.

$\frac{w_1}{w_2} \rightarrow$ rational



$$r_{i+1} = r_i$$
$$\theta_{i+1} = \theta_i + 2\pi \frac{w_1}{w_2}$$

$$\frac{w_1}{w_2} = \frac{\frac{\partial H_0(J_1, J_2)}{\partial J_1}}{\frac{\partial H_0(J_1, J_2)}{\partial J_2}} = f(J_1, J_2)$$

Also the system is conservative,

$$H_0(J_1, J_2) = E$$

$$\frac{w_1}{w_2} = a(r)$$

Then

$$r_{i+1} = r_i$$
$$\theta_{i+1} = \theta_i + 2\pi a(r)$$

map is area preserving.

$a(r) = \frac{w_1}{w_2} = n/s$ $\qquad$ n, s co-prime

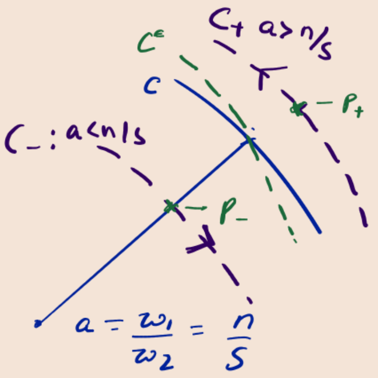
$$T^s \begin{pmatrix} r_0 \\ \theta_0 \end{pmatrix} = \begin{bmatrix} r_0 \\ \theta_0 + 2\pi \frac{n}{s} \cdot s \end{bmatrix} = \begin{bmatrix} r_0 \\ \theta_0 \end{bmatrix}$$

r_0, θ_0 are fixed points of the S^{th} iterate.

Now let's add a perturbation.

$$r_{i+1} = r_i + \epsilon f(r_i, \theta_i)$$
$$\theta_{i+1} = \theta_i + 2\pi a(r_i) + \epsilon g(r_i, \theta_i)$$

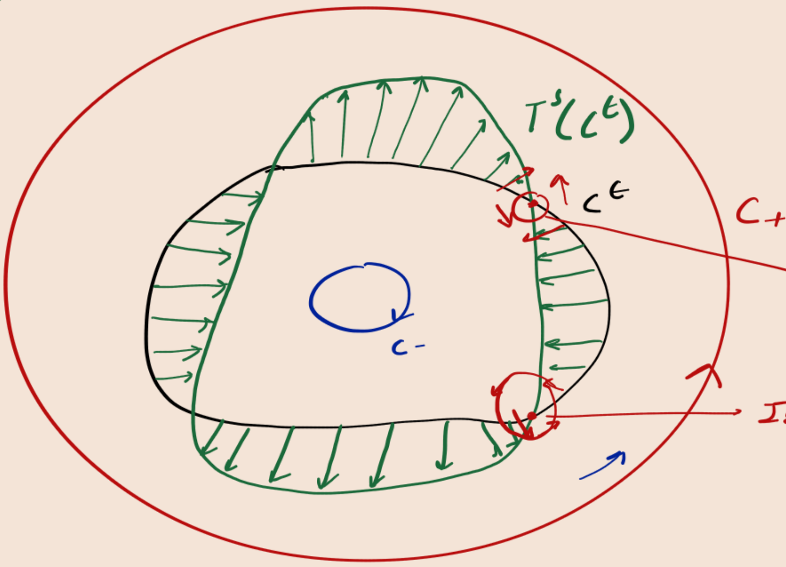
f, g should be such perturbations such that the perturbed map is also conservative.



→ Let C be a curve on the surface of the torus such that all points on C are fixed points for the s^{th} iterate of the unperturbed map.

Note the points P_+ and P_- are not fixed points of the s^{th} iterate. On taking s iterations P_+^s will move a bit further than P_+ and P_-^s will move a bit lesser than P_-.

C^ϵ is our perturbed curve. C^ϵ and C must cross an even number of times since the map has to be area preserving.



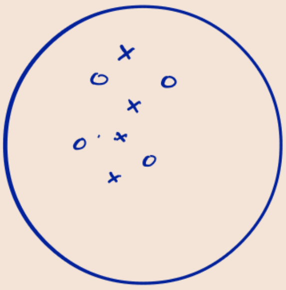
The green curve is the s^{th} iterate of the perturbed curve C^ϵ.

This is a saddle.

Is an elliptic fixed point.

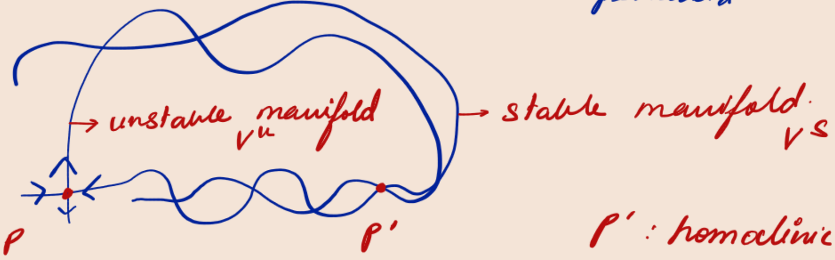
The fixed points alternate between saddles and elliptic fixed points

The mechanism of Tori-Breaking is called Poincaré Birkhoff theorem. (Read more on this)



× → saddle   ○ → elliptic fp

We cannot have stable homoclinic bifurcations.
Under perturbation,



p' : homoclinic pt.

$p \in V^u$, $p \in V^s$

Let's take the map $f^n(p')$

$$\lim_{n\to\infty} f^n(p') \to p \qquad p' \text{ in the stable direction.}$$

inverse of the map

$$\lim_{n\to\infty} f^{-n}(p') \to p \qquad p' \text{ in the unstable direction.}$$

As $n \to \infty$

$$f^n(f(p')) \to p$$

$$f^{-n}(f(p')) \to p$$

Thus we can see that $f(p')$ is also a homo-clinic point.

Similarly,

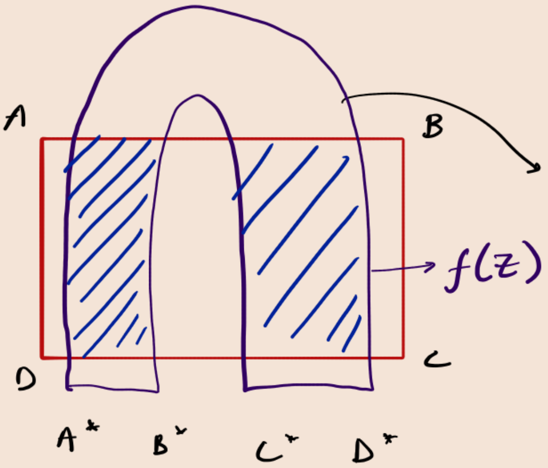
$$f^n(f^{-1}(p')) \to p$$

$$f^{-n}(f^{-1}(p')) \to p$$

Any forward iterate or backward iterate will also be a homoclinic pt.

(homoclinic tangle)

⊛ For chaos to show, it must be on a fractal set.
If you have dynamics of streching and folding (like the Baker Map), it is in general called a (if it is area conservings) Smale Horsshoe map.

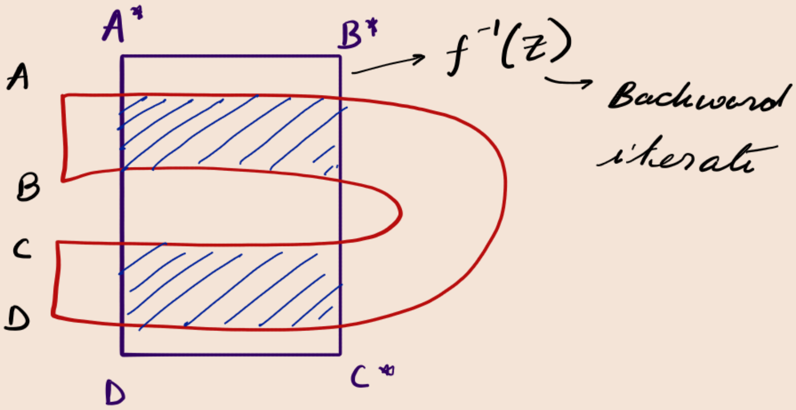


$Ar(ABCD) = Ar(A^* B^* C^* D^*)$

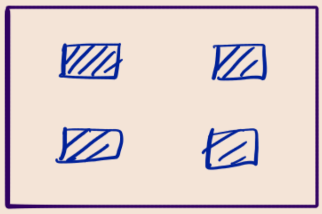
forward iterate of the set of initial condition ABCD.

Let Z = region ABCD

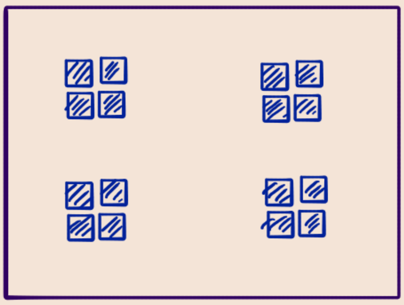
If we start out with $A^* B^* C^* D^*$ as our initial conditions., we get



$f(Z) \cap Z \cap f^{-1}(Z)$

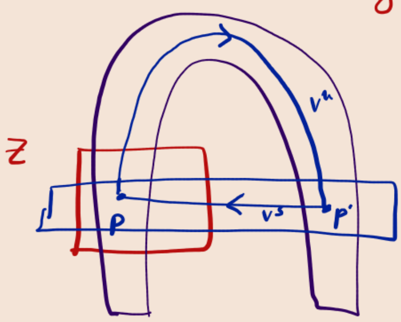


Now if we take $f^2(Z) \cap Z \cap f^{-2}(Z)$



Appearance of Fractal-like sets

(x) This smale-Horseshoe can also be related to the Homoclinic tangle.



Homoclinic tangle. Non wandering set.

References = Schüster